

7.1 Area Btwn 2 Curves:

$$A = \int_a^b f(x) - g(x) dx$$

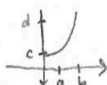
↑ top - bottom

$$A = \int_a^b [f(x) - g(x)] dx + \int_b^c [g(x) - f(x)] dx$$

7.2 Disk & Washer

$$V_{HA} = \pi \int_a^b [f(x)]^2 dx$$

$$V_{VA} = \pi \int_c^d [g(y)]^2 dy$$



$$\text{For } y=n: V_{HA} = \int_a^b [F(x) - n]^2 dx$$

$$\text{Washer: } V_{HA} = \pi \int_a^b [f(x)^2 - g(x)^2] dx$$

$$V_{VA} = \pi \int_c^d [F(y)^2 - G(y)^2] dy$$

outside R = top - axis of rot.
inside R = bot - axis of rot

7.3 Shell

for VA/donut convert to y / one curve

$$V_{VA} = 2\pi \int_a^b f(x) \cdot x dx; V_{HA} = 2\pi \int_c^d g(y) \cdot y dy$$

$$\text{Detail: } 2\pi \int_a^b (\text{top-bot}) \cdot (x\text{-axis}) dx$$

7.4 Arc length & Surface

$$y = f(x) = \int_a^b \sqrt{1 + [f'(x)]^2} dx \quad \text{Arc length}$$

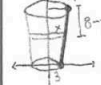
$$x = g(y) = \int_c^d \sqrt{1 + [g'(y)]^2} dy \quad \text{Arc length}$$

$$\text{Surface: } SA_{HA} = \int_a^b 2\pi f(x) \cdot \sqrt{1 + [f'(x)]^2} dx$$

$$SA_{VA} = \int_c^d 2\pi g(y) \cdot \sqrt{1 + [g'(y)]^2} dy$$

7.5 Cont.

4) Work req. to pump water out a tank; water = 62.4 lb/ft³



$$W = F \cdot D = \text{Weight density} \times \text{Volume} \times \text{distance}$$

$$= \int_0^8 62.4 \cdot \pi x^2 (8-y) dy$$

change to y w/ line

7.5 Work (W)

W = constant force × distance

$$W = F \cdot D$$

By variable force: $W = \int_a^b F(x) dx$

Spring Examples

Hooke's Law: $F(x) = kx$; k is distance

1) Force of 10 lbs to hold spring stretched 4ft beyond. How much work for natural to 6ft?
 $F(x) = kx$
 $F(4) = 10 = 4k \Rightarrow k = \frac{5}{2} \Rightarrow W = \int_0^6 \frac{5}{2} x dx = 45 \text{ ft-lb}$

2) 2J of work for natural 10cm to 12cm.
Force to stretch to 12cm?

$$W = 2J \quad W = \int_0^{0.12} F(x) dx = 2 \quad k = \frac{2}{\frac{1}{2} (0.12)^2} = 20000$$

3) Lifting: 15lb object up 30-ft building if cable weights 1.5 lbs/ft

$$W = 15 \text{ ft-lb} \quad W = \int_0^{30} F(x) dx = 15(30) + \frac{1}{2} (1.5)(30)^2 = 450 \text{ ft-lb}$$

7.6 Moment, Center of mass, centroid

F = ma

Moment = mass × distance

$$\bar{x} = \frac{\text{total moments}}{\text{total mass}} = \frac{m_1 x_1 + m_2 x_2 + \dots}{m_1 + m_2 + \dots}$$

Center of mass = center of gravity

Center of mass = $(\bar{x} = \frac{M_y}{M_T}, \bar{y} = \frac{M_x}{M_T})$; M_T = total mass

CoM Planar Lamina (centroid)

$$M_y = \rho \int_a^b (f(x) - g(x)) x dx; \bar{x} = \frac{M_y}{M_T} = 0 \text{ (when sym)}$$

$$M_x = \rho \int_a^b \frac{(f(x)^2 - g(x)^2)}{2} dx; \bar{y} = \frac{M_x}{M_T}$$

$$M_T = \rho \int_a^b [f(x) - g(x)] dx$$

8.1 Basic Integration

Power rule: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$; $n \neq -1$

Log: $\int \frac{1}{x} dx = \ln|x| + C$

$$\text{w/o } u: \int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$$

nat. Log: $\int e^x dx = e^x + C$

$$\text{w/o } u: \int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

0 $\int 2xe^{-x^2} dx \rightarrow \text{let } u = -x^2$

8.2 Improper Integral

$$\int_1^\infty \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1} & p > 1, \text{ conv.} \\ \infty & p \leq 1, \text{ div.} \end{cases}$$

Infinite integration:

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx \quad \text{lim exist = conv.}$$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx \quad \text{lim DNE = div.}$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx \quad \text{right exists = conv.}$$

Infinite discontinuities

$$[a, b) \rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

$$(a, b] \rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

$$[a, c) \cup (c, b] \rightarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$= \lim_{t \rightarrow c^-} \int_a^t f(x) dx + \lim_{t \rightarrow c^+} \int_t^b f(x) dx$$

$$\int \sin^2 x \cos x dx \rightarrow \text{let } u = \sin x, du = \cos x dx$$

$$\int \sin^2 x \cos x dx \rightarrow \text{let } u = \sin x$$

$$\int \tan^2(x) \sec^2(x) dx \rightarrow \text{let } u = \tan(x)$$

$$du = \sec^2(x) dx \rightarrow \frac{1}{2} du = \sec^2(x) dx$$

$$\int \cos^2(x) \sin(x) dx \rightarrow \text{let } u = \cos(x)$$

$$du = -\sin(x) dx \rightarrow -\frac{1}{2} du = \sin(x) dx$$

$$\int \sec^2(x) \tan^2(x) dx \rightarrow u = \sec x, du = \sec x \tan x dx$$

$$\int \sec^2(x) \tan^2(x) dx \rightarrow \sec^2(x) \tan^2(x) dx$$

$$\int \sin^2(x) \cos^2(x) dx \rightarrow \sin^2(x) \cos^2(x) dx$$

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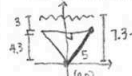
7.7 Fluid Force & Pressure

Pressure = weight density × depth = $P = wh$

$F = P \times \text{Area}$

Irregular shape: $F = \int_a^b h(y) \cdot 2\pi y dy$

1) Equilateral Δ (6ft); 3ft under water; fluid force = ?



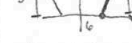
$$\text{rewrite } x: (0,0), (2,5), (4,0) \Rightarrow F = \int_0^3 62.4 (7.3-y) 2x dy$$

$$C = \sqrt{5^2 - 2^2} = 4.3$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-0}{2-0} = \frac{5}{2}$$

$$y - y_1 = m(x - x_1) \Rightarrow y = \frac{5}{2}x$$

2) Trapezoid



$$\text{rewrite } x: F = 62.4 \int_4^5 (9-y) 2x dy$$

4) semi-circle



$$\text{convert units } x^2 + y^2 = r^2 \Rightarrow x = \sqrt{0.6^2 - y^2} \Rightarrow F = 9800 \int_0^{0.6} (2.75-y) 2x dy$$

8.1 Basic Integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C; n \neq -1$$

$$\log: \int \frac{1}{x} dx = \ln|x| + C$$

$$\text{w/o } u: \int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$$

nat log: $\int e^x dx = e^x + C$

$$\text{w/o } u: \int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

8.2 Integration By Parts

i.e. $\int x^2 dx, \int x \ln x, \int x \sin x$, etc

$$\int u dv = uv - \int v du; \text{ let } u = f(x), dv = g(x) dx$$

$$du = f'(x) dx, v = g(x)$$

LIATE: Log, Inv trig, Alg, Trig, exp

Tabular for $\int x^n \text{trig}, \int x^n e^x, \int \text{sign, den, inte.}$

$$\int x^2 \ln x dx \rightarrow uv - \int v du$$

$$\text{let } u = \ln x, dv = x^2 dx \rightarrow \frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx$$

$$du = \frac{1}{x} dx, v = \frac{x^3}{3} \rightarrow \frac{x^3}{3} \ln x - \frac{1}{4} x^4 + C$$

$$\int x^2 \sin 2x dx \rightarrow uv - \int v du$$

$$\text{let } u = \sin 2x, dv = x^2 dx \rightarrow \frac{x^3}{3} \sin 2x - \int \frac{x^3}{3} \cdot 2 \cos 2x dx$$

$$du = 2 \cos 2x, v = \frac{x^3}{3} \rightarrow \frac{x^3}{3} \sin 2x - \frac{2}{3} \int x^3 \cos 2x dx$$

$$= \frac{1}{5} (e^x \sin x - 2e^x \cos x) + C$$

8.3 Trig Sub

$\sqrt{a^2 - u^2}$, let $u = a \sin \theta$ # B-16

$\sqrt{a^2 + u^2}$, let $u = a \tan \theta$

$\sqrt{u^2 - a^2}$, let $u = a \sec \theta$

$\frac{u}{\sqrt{a^2 - u^2}}$, let $u = a \sin \theta$

$\frac{u}{\sqrt{a^2 + u^2}}$, let $u = a \tan \theta$

$\frac{u}{\sqrt{u^2 - a^2}}$, let $u = a \sec \theta$

$\frac{1}{\sqrt{a^2 - u^2}}$, let $u = a \sin \theta$

$\frac{1}{\sqrt{a^2 + u^2}}$, let $u = a \tan \theta$

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$\frac{1}{\sqrt{a^2 + u^2}}$, let $u = a \tan \theta$

$\frac{1}{\sqrt{u^2 - a^2}}$, let $u = a \sec \theta$

8.3 Trig Integral

$\sin^\circ$ = odd, $\oplus \rightarrow$ save $\sin x$, convert rest to $\cos$

$\cos^\circ$ = odd, $\oplus \rightarrow$ save $\cos x$, convert rest to $\sin$

$\sin^2 x + \cos^2 x = 1 \rightarrow 1 - \cos^2 x = \sin^2 x \rightarrow \cos^2 x = 1 - \sin^2 x$

$\sin^\circ, \cos^\circ$ = even, $\oplus \rightarrow$ reduction formula

reduction: $\sin^2 x = \frac{1 - \cos 2x}{2}, \cos^2 x = \frac{1 + \cos 2x}{2}$

also: $\int \cos x dx = \frac{1}{a} \sin ax + C, \sin ax = 2 \sin x \cos x$

$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \frac{1}{2} \sin 2x = \sin x \cos x$

$\sec^\circ$ = even, $\oplus \rightarrow$ save $\sec^2 x$, convert rest to $\tan$

$\frac{d}{dx} \tan x = \sec^2 x dx; 1 + \tan^2 x = \sec^2 x, \tan^2 x = \sec^2 x - 1$

$\tan^\circ$ = odd, $\oplus \rightarrow$ save $\sec x \tan x$, convert rest to $\sec$

$\frac{d}{dx} \sec x = \sec x \tan x dx$

$\tan^\circ$ = even, $\oplus$, w/o $\sec \rightarrow$ convert $\tan^2 x$ to $\sec^2 x - 1$

Handwritten notes on calculus topics including integration, sequences, and series. The notes are organized into sections with various mathematical formulas and derivations.

8.6 Numerical Integration

- $\Delta x = \frac{b-a}{n}$
- Trapezoid Rule: $\frac{\Delta x}{2} [f(a) + 2f(x_1) + 2f(x_2) + \dots + f(b)]$
- Simpson's Rule: $\frac{\Delta x}{3} [f(a) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + f(b)]$
- Midpoint: $\Delta x [f(m_1) + f(m_2) + \dots + f(\frac{x_k + x_{k+1}}{2})]$
- Error Bounds
 - Error in $M_n \leq \frac{M(b-a)^2}{24n^2}$, M is max of $|f''(x)|$ over $[a, b]$
 - Error in $T_n \leq \frac{M(b-a)^3}{12n^2}$
 - Error in $S_n \leq \frac{M(b-a)^5}{180n^4}$, M is max of $|f^{(4)}(x)|$

8.7 Int By Partial Fraction

- for polynomial/polynomial
- If top > bot $\rightarrow$ long division & decompose
- If top < bot $\rightarrow$ decompose
- to decompose: Factor, LCD, plug x, rewrite
- then integrate: $\int \frac{1}{a^2+u^2} du = \frac{1}{a} \arctan \frac{u}{a} + C$ or #17 $\int \frac{1}{a^2-u^2} du$

9.1 Sequence, converge/diverge

- SEQ > 2nd > STAT > OPS > 5 seq > ... > ENTER
- Find a_n : $(-1)^n$ when $a = 0$, $(-1)^{n+1}$ when $a = 0$
- Factorial: $n! = n(n-1)(n-2) \dots$
- Arithmetic: $a_n = a_1 + (n-1)d$, d = difference
- Geometric: $a_n = a_1 \cdot r^{n-1}$, r = common ratio
- Con/Diverge: $\lim_{n \rightarrow \infty} a_n = L$
- Arithmetic = div; geometric = div. when $|r| > 1$

9.2 Series

- Partial sum: $S_n = \sum_{k=1}^n a_k$
- Conv/Div
 - Find partial frac
 - Partial sum w/ $n=1, 2, 3, n-2, n-1, n$
 - $\{s_n\}$ converge to $S \rightarrow \sum_{n=1}^{\infty} a_n = S$
- Geometric Series
 - If $|r| \geq 1 \rightarrow$ diverge
 - If $|r| < 1 \rightarrow$ converge to $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$
- N^{th} Term Test for Divergence
 - only if $\sum_{n=1}^{\infty} a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$
 - If $\lim_{n \rightarrow \infty} a_n \neq 0$, $\sum_{n=1}^{\infty} a_n$ diverges

9.3 Integral Test

- Conditions $a_n = f(n)$
 - $f(n) > 0$, continuous, decrease for all $n \geq 1$
 - Graph a_n to check $a_n > 0$ interval decrease
 - $\sum_{n=1}^{\infty} a_n \rightarrow \int_1^{\infty} f(x) dx \rightarrow \lim_{x \rightarrow \infty} f(x) = \text{conv. or div.}$
- P-Series Test
 - $\int_1^{\infty} \frac{1}{x^p} dx \rightarrow$ conv. if $p > 1$, div. if $p \leq 1$
- (Integral test) $\int_1^{\infty} \frac{1}{x^p} dx \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^p}$ conv. if $p > 1$
- (N^{th} Term Div. Test)
 - $\sum_{n=1}^{\infty} \frac{1}{n^p} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ if $p > 0$ diverge if $p \leq 0$
- (Int. Test) Harmonic
 - $\int_1^{\infty} \frac{1}{x} dx \rightarrow \sum_{n=1}^{\infty} \frac{1}{n}$ diverge
 - $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ alt. harm. $\rightarrow$ conv

9.4 Comparison Test

- Direct Comp: $0 < a_n < b_n$, a_n (smaller), b_n (bigger)
 - If $\sum_{n=1}^{\infty} b_n$ converge, then $\sum_{n=1}^{\infty} a_n$ converge
 - If $\sum_{n=1}^{\infty} a_n$ diverge, then $\sum_{n=1}^{\infty} b_n$ diverge
 - If b_n diverge $\rightarrow$ can't conclude
- Limit Comp.
 - If $a_n > 0$, $b_n > 0$, and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$, (L finite, $\neq 0$)
 - then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both conv./div.
 - Tip: b_n = (easier highest) for poly/polynomial

9.5 Alternating Series

- $\sum_{n=1}^{\infty} (-1)^n a_n$ or $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges when $a_n > 0$
- if $\lim_{n \rightarrow \infty} a_n = 0$ & $a_n \leq a_{n-1}$ for all n
- Absolute conv: If $\sum |a_n|$ converge, then $\sum a_n$ converge
- Conditionally conv: If $\sum a_n$ conv., but $\sum |a_n|$ diverge
- Approximation Error
 - remainder: $R_N = S - S_N$, $|S - S_N| = |R_N| \leq a_{N+1}$

9.6 Ratio & Root Test

- Ratio Test
 - $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \rightarrow < 1$, converge absolutely
 - $\rightarrow > 1$ or ∞ , $\sum a_n$ diverge
 - $\rightarrow 1$, inconclusive
 - guarantee alternating sign version
 - for exponential/factorial
- Root Test
 - $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \rightarrow < 1$, $\sum a_n$ conv. absolutely
 - $\rightarrow > 1$ or ∞ , diverge
 - $\rightarrow 1$, inconclusive
- for series raised to power "n"

9.7 Taylor Polynomials & Approximation

- N^{th} Taylor Polynomial
 - If f has N derivative at $x=c$, then
 - $P_N(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots + \frac{f^{(N)}(c)}{N!}(x-c)^N$
- N^{th} Maclaurin Polynomial
 - If f has N derivative at $x=0$, then
 - $P_N(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(N)}(0)}{N!}x^N$
- Power series: x is variable
 - $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$ center at 0
 - If series converges at specific domain, then sum of series will converge to output of function f at x
 - half of interval of convergence = radius

9.8 Power Series

- Some functions can be represented exactly by an infinite series called power series
 - $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ sum of Taylor polynomial
- $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$ or $\sum_{n=0}^{\infty} a_n x^n$, where "x" so converge
- Power series: x is variable
 - $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$ center at 0
 - If series converges at specific domain, then sum of series will converge to output of function f at x
 - half of interval of convergence = radius

9.9 Represent Functions By Power Series

- Power Series general form: $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$
- Derivative: $f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} (x-c)^n$
- Integral: $\int f(x) dx = \sum_{n=0}^{\infty} \frac{a_n (x-c)^{n+1}}{n+1} + C$
- radius of convergence same
- interval of convergence may differ b/c endpoint behavior

9.10 Taylor & Maclaurin Series

- If f represented by $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$
 - $a_n = \frac{f^{(n)}(c)}{n!}$
 - $f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots + \frac{f^{(n)}(c)}{n!}(x-c)^n$
- If function f has derivative of all orders at $x=c$
 - $\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(x)$
 - Taylor series for f at c
 - Maclaurin series for $c=0$
- Bi-modal series: Maclaurin series for $f(x) = \ln(x)$
 - $\ln(x) = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

8.8 Improper Integral

- Infinite Interval
 - $\int_a^{\infty} \frac{1}{x^p} dx = \left[p > 1 \rightarrow \text{convergent}, \frac{1}{p-1} \right] \left[p \leq 1 \rightarrow \text{divergent} \right]$
- Infinite Integration
 - $\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$ limit exists = converge
 - $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$ limit DNE = diverge
 - $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$ exist = converge
- Infinite Discontinuities
 - $[a, b] \rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$
 - $(a, b] \rightarrow \int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$
 - $[a, c), (c, b] \rightarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$
 - $\lim_{t \rightarrow c^-} \left[\int_a^t f(x) dx \right] + \lim_{t \rightarrow c^+} \left[\int_t^b f(x) dx \right]$

EC: Monotonic

- $a_n \leq a_{n+1}$ for all n , (increasing)
- $a_n \geq a_{n+1}$ for all n , (decreasing)
- Note: $(2n+1)!$

8.6 Trig Sub

- $\sqrt{a^2 - u^2}$, Let $u = a \sin \theta$
- $\sqrt{a^2 + u^2}$, Let $u = a \tan \theta$
- $\sqrt{u^2 - a^2}$, Let $u = a \sec \theta$
- table #13-16

8.6 Numerical Integration

- $\Delta x = \frac{b-a}{n}$
- Trapezoid Rule: $\frac{\Delta x}{2} [f(a) + 2f(x_1) + 2f(x_2) + \dots + f(b)]$
- Simpson's Rule: $\frac{\Delta x}{3} [f(a) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + f(b)]$
- Midpoint: $\Delta x [f(m_1) + f(m_2) + \dots + f(\frac{x_k + x_{k+1}}{2})]$
- Error Bounds
 - Error in $M_n \leq \frac{M(b-a)^2}{24n^2}$, M is max of $|f''(x)|$ over $[a, b]$
 - Error in $T_n \leq \frac{M(b-a)^3}{12n^2}$
 - Error in $S_n \leq \frac{M(b-a)^5}{180n^4}$, M is max of $|f^{(4)}(x)|$

9.1 Sequence, converge/diverge

- SEQ > 2nd > STAT > OPS > 5 seq > ... > ENTER
- Find a_n : $(-1)^n$ when $a = 0$, $(-1)^{n+1}$ when $a = 0$
- Factorial: $n! = n(n-1)(n-2) \dots$
- Arithmetic: $a_n = a_1 + (n-1)d$, d = difference
- Geometric: $a_n = a_1 \cdot r^{n-1}$, r = common ratio
- Con/Diverge: $\lim_{n \rightarrow \infty} a_n = L$
- Arithmetic = div; geometric = div. when $|r| > 1$

9.2 Series

- Partial sum: $S_n = \sum_{k=1}^n a_k$
- Conv/Div
 - Find partial frac
 - Partial sum w/ $n=1, 2, 3, n-2, n-1, n$
 - $\{s_n\}$ converge to $S \rightarrow \sum_{n=1}^{\infty} a_n = S$
- Geometric Series
 - If $|r| \geq 1 \rightarrow$ diverge
 - If $|r| < 1 \rightarrow$ converge to $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$
- N^{th} Term Test for Divergence
 - only if $\sum_{n=1}^{\infty} a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$
 - If $\lim_{n \rightarrow \infty} a_n \neq 0$, $\sum_{n=1}^{\infty} a_n$ diverges

9.3 Integral Test

- Conditions $a_n = f(n)$
 - $f(n) > 0$, continuous, decrease for all $n \geq 1$
 - Graph a_n to check $a_n > 0$ interval decrease
 - $\sum_{n=1}^{\infty} a_n \rightarrow \int_1^{\infty} f(x) dx \rightarrow \lim_{x \rightarrow \infty} f(x) = \text{conv. or div.}$
- P-Series Test
 - $\int_1^{\infty} \frac{1}{x^p} dx \rightarrow$ conv. if $p > 1$, div. if $p \leq 1$
- (Integral test) $\int_1^{\infty} \frac{1}{x^p} dx \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^p}$ conv. if $p > 1$
- (N^{th} Term Div. Test)
 - $\sum_{n=1}^{\infty} \frac{1}{n^p} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ if $p > 0$ diverge if $p \leq 0$
- (Int. Test) Harmonic
 - $\int_1^{\infty} \frac{1}{x} dx \rightarrow \sum_{n=1}^{\infty} \frac{1}{n}$ diverge
 - $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ alt. harm. $\rightarrow$ conv

9.4 Comparison Test

- Direct Comp: $0 < a_n < b_n$, a_n (smaller), b_n (bigger)
 - If $\sum_{n=1}^{\infty} b_n$ converge, then $\sum_{n=1}^{\infty} a_n$ converge
 - If $\sum_{n=1}^{\infty} a_n$ diverge, then $\sum_{n=1}^{\infty} b_n$ diverge
 - If b_n diverge $\rightarrow$ can't conclude
- Limit Comp.
 - If $a_n > 0$, $b_n > 0$, and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$, (L finite, $\neq 0$)
 - then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both conv./div.
 - Tip: b_n = (easier highest) for poly/polynomial

9.5 Alternating Series

- $\sum_{n=1}^{\infty} (-1)^n a_n$ or $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges when $a_n > 0$
- if $\lim_{n \rightarrow \infty} a_n = 0$ & $a_n \leq a_{n-1}$ for all n
- Absolute conv: If $\sum |a_n|$ converge, then $\sum a_n$ converge
- Conditionally conv: If $\sum a_n$ conv., but $\sum |a_n|$ diverge
- Approximation Error
 - remainder: $R_N = S - S_N$, $|S - S_N| = |R_N| \leq a_{N+1}$

9.6 Ratio & Root Test

- Ratio Test
 - $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \rightarrow < 1$, converge absolutely
 - $\rightarrow > 1$ or ∞ , $\sum a_n$ diverge
 - $\rightarrow 1$, inconclusive
 - guarantee alternating sign version
 - for exponential/factorial
- Root Test
 - $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \rightarrow < 1$, $\sum a_n$ conv. absolutely
 - $\rightarrow > 1$ or ∞ , diverge
 - $\rightarrow 1$, inconclusive
- for series raised to power "n"

9.7 Taylor Polynomials & Approximation

- N^{th} Taylor Polynomial
 - If f has N derivative at $x=c$, then
 - $P_N(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots + \frac{f^{(N)}(c)}{N!}(x-c)^N$
- N^{th} Maclaurin Polynomial
 - If f has N derivative at $x=0$, then
 - $P_N(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(N)}(0)}{N!}x^N$
- Power series: x is variable
 - $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$ center at 0
 - If series converges at specific domain, then sum of series will converge to output of function f at x
 - half of interval of convergence = radius

9.8 Power Series

- Some functions can be represented exactly by an infinite series called power series
 - $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ sum of Taylor polynomial
- $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$ or $\sum_{n=0}^{\infty} a_n x^n$, where "x" so converge
- Power series: x is variable
 - $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$ center at 0
 - If series converges at specific domain, then sum of series will converge to output of function f at x
 - half of interval of convergence = radius

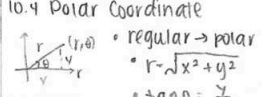
9.9 Represent Functions By Power Series

- Power Series general form: $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$
- Derivative: $f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} (x-c)^n$
- Integral: $\int f(x) dx = \sum_{n=0}^{\infty} \frac{a_n (x-c)^{n+1}}{n+1} + C$
- radius of convergence same
- interval of convergence may differ b/c endpoint behavior

9.10 Taylor & Maclaurin Series

- If f represented by $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$
 - $a_n = \frac{f^{(n)}(c)}{n!}$
 - $f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots + \frac{f^{(n)}(c)}{n!}(x-c)^n$
- If function f has derivative of all orders at $x=c$
 - $\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(x)$
 - Taylor series for f at c
 - Maclaurin series for $c=0$
- Bi-modal series: Maclaurin series for $f(x) = \ln(x)$
 - $\ln(x) = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

10.1
 • Ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
 • Hyperbola: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$



- 10.2 Parametric Eq. (Pt.)
 • Sketch curve by RE
 • $\frac{dx}{dt}$ & $\frac{dy}{dt}$, orientation arrow
 • Eliminate Parameter:
 • $x = e^{2t}$ $y = e^{2t}$
 $\rightarrow t = \frac{1}{2} \ln x$ $y = e^{2(\frac{1}{2} \ln x)} = x$
 • $\sin^2 \theta + \cos^2 \theta = 1$
 • $x = 3 \cos \theta \rightarrow \cos \theta = \frac{x}{3}$
 • $y = 4 \sin \theta \rightarrow \sin \theta = \frac{y}{4}$ $(\frac{x}{3})^2 + (\frac{y}{4})^2 = 1$

- 10.3 Slope/Tangent Lines, Arc/SA
 • Slope/Tan/Derivative of Curve
 • $x = f(t)$ & $y = g(t)$, slope c @ (x, y)
 • $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, where $\frac{dx}{dt} \neq 0$ = slope
 • Concavity: $\frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right] = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{\frac{dx}{dt}}$

- Vertical/Horizontal Tangent
 • Prolate: find slope w/ $\frac{y'}{x'}$
 • then plug in points to get y & t
 • solve for slope at T
 • plug into $y = mx + b$

- Polar curve: $r = e \cos \theta$
 • $x = r \cos \theta = e \cos^2 \theta \rightarrow x' = -8 \sin 2\theta$
 • $y = r \sin \theta = e \cos \theta \sin \theta = e \cos 2\theta$
 • plug $x' = 0$ to get θ & solve (x, y) - vert
 • plug $y' = 0$ for θ & solve (x, y) - horiz
 • Arc length: $S = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$
 • SA abt x-axis: $S = 2\pi \int_a^b g(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$, $y = g(t)$
 abt y-axis: $S = 2\pi \int_a^b f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$, $x = f(t)$

- 10.4 Polar Coordinate
 • regular $\rightarrow$ polar
 • $r = \sqrt{x^2 + y^2}$
 • $\tan \theta = \frac{y}{x}$
 • $\theta = \tan^{-1}(\frac{y}{x})$
 • rose petal
 • $r = a \sin n\theta$
 • 1st petal: $\frac{\pi}{2n}$
 • $r = a \cos n\theta$
 • 1st petal: 0
 • $a > 0$
 • $n = \text{odd} \rightarrow n$ petals
 • $n = \text{even} \rightarrow 2n$ petals
 • polar $\rightarrow$ regular
 • $\cos \theta = \frac{x}{r} \rightarrow x = r \cos \theta$
 • $\sin \theta = \frac{y}{r} \rightarrow y = r \sin \theta$
 • $(r, \theta) = (r, \theta + 2k\pi)$
 = $(-r, \theta + (2k+1)\pi)$ $\rightarrow$ add π to check

- Slope & Tangent Line
 • $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} \rightarrow 0$, horizontal tan.
 $\frac{dy}{d\theta} = 0$, vertical tan.

- ① Graph: $r = 2 \cos \theta$
 $r = r \cos \theta$
 $x^2 + y^2 = 2x \rightarrow x^2 - 2x + y^2 = 0$
 $\rightarrow$ complete sq.

- ② $r = 8 \cos \theta$, horiz. tan.
 $y = r \sin \theta \rightarrow y = 8 \cos \theta \sin \theta$
 $y = 4 \sin 2\theta$
 $\frac{dy}{d\theta} = 8 \cos 2\theta = 0 \rightarrow \cos 2\theta = 0$
 $2\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
 $r = 8 \cos(\frac{\pi}{4}) = 4\sqrt{2}$
 $\rightarrow (4\sqrt{2}, \frac{\pi}{4})$

- Application: growth & decay
 • $\frac{dy}{dt} = ky$, k = proportional c.
 $\int \frac{1}{y} dy = \int k dt$
 $\ln |y| = kt + c$
 $y = \pm e^{kt+c}$
 $y = Ce^{kt}$
 • $k > 0$: grow
 • $k < 0$: decay
 • C : initial value

- Newton's Cooling Law
 • $\frac{dT}{dt} = k(T - T_s)$
 • $T(t)$ funct. & $T_s = \text{temp. surrounding}$

- 6.1 Slope Fields
 • Differential Equation Solutions
 • general solution: $y = (e^x) \cdot C$
 • Particular solution at $(0, 2)$
 • $y(0) = 2 \rightarrow 2 = (e^0) \cdot C \rightarrow C = 2$
 $\rightarrow y = 2e^x$
 • Slope Fields
 • draw slope at each point
 • $\frac{dy}{dx}$ to approximate sol.

- 6.2-3 Separation Variables
 • Separation of Variables
 • Solve general DE, sep. thru multipln
 • ex: $\frac{dy}{dx} = xy$ or $(x^2+1)(y-1)$ or $(xy-x = x(y-1))$
 • $\frac{dy}{dx} = f(x)g(y)$
 $\rightarrow \int \frac{1}{g(y)} dy = \int f(x) dx$ integrate to solve y
 $\rightarrow \int g(y) dy = \int f(x) dx$ check $f \cdot g(y) = 0$ p sol.
 • one big C

- ① Rate of change of pop
 $P = Ce^{kt}$ $t=0, 100$ $t=5, 50$ pop $t=10, ?$ pop, $P(0)=100$
 $t=0, 100 = (e^0) \rightarrow 100 = P = 100e^k$
 $t=5, 50 = 100e^{5k}$
 $\frac{1}{2} = e^{5k} \rightarrow \ln(\frac{1}{2}) = 5k \rightarrow k = \frac{\ln(\frac{1}{2})}{5} = -\frac{\ln 2}{5}$
 $P(t) = 100e^{-\frac{\ln 2}{5}t} \rightarrow P(10) = 25$

- ② Sales drop from 100k/month $\rightarrow$ 80k/month, after 2 months, $t=4m$
 $y = Ce^{kt}$ $t=0 \rightarrow 100k$, $t=4 \rightarrow 80k$, $t=?$
 $100 = Ce^0 \rightarrow C = 100k$
 $t=4, 80 = 100e^{4k}$
 $\frac{80}{100} = e^{4k} \rightarrow 4k = \ln \frac{80}{100}$
 $k = -0.055$
 $y = 100e^{-0.055t}$
 $\rightarrow y(6) = 71k$

- ① $y = 3e^{2x}$ solution of $y' + 2y = 0$
 $y' = 6e^{2x}$
 $y' + 2y = 6e^{2x} + 2(3e^{2x}) = 6e^{2x} + 6e^{2x} = 12e^{2x} \neq 0$
 ② $y = Ce^{-t}$ solution of $y' = y-1$ w/ $y(0) = 5$
 $y' = -Ce^{-t} \rightarrow -Ce^{-t} = Ce^{-t} - 1 \rightarrow 1 = 2Ce^{-t}$
 $y(0) = 5 \rightarrow (0, 5) \rightarrow 5 = (e^0) \cdot C \rightarrow C = 5$
 $y = 5e^{-t}$
 $C = 6 \rightarrow y = 6e^{-t} - 1$ = Particular sol

- ① General solution: $\frac{dy}{dx} = (x^2 - 4)(3y + 2)$
 $\int \frac{1}{3y+2} dy = \int (x^2 - 4) dx$
 $\frac{1}{3} \ln |3y+2| = \frac{x^3}{3} - 4x + C$
 $\ln |3y+2| = x^3 - 12x + C$
 $3y+2 = e^{x^3 - 12x + C} = e^{x^3 - 12x} \cdot e^C$
 $3y+2 = Ce^{x^3 - 12x}$
 $y = \frac{Ce^{x^3 - 12x} - 2}{3}$
 $3y - 2 = 0 \rightarrow y = \frac{2}{3}$
 ② $y' = 2xy + 3y - 4x - 6$, $y(0) = 5$
 $\frac{dy}{dx} = (y-2)(2x+3)$
 $\int \frac{1}{y-2} dy = \int (2x+3) dx$
 $\ln |y-2| = x^2 + 3x + C$
 $y-2 = e^{x^2 + 3x + C} = e^{x^2 + 3x} \cdot e^C$
 $y-2 = Ce^{x^2 + 3x}$
 $y = Ce^{x^2 + 3x} + 2$
 $y(0) = 5 \rightarrow (0, 5) \rightarrow 5 = C + 2 \rightarrow C = 3$
 $y = 3e^{x^2 + 3x} + 2$

- ③ Pizza cool from 350° , kitchen = $75^\circ F$.
 5 mins later, pizza 340° . How much until $300^\circ F$?
 Newton's cooling: $T(t) = 350 \rightarrow 350 = (e^0) \cdot 15 + 75$
 $\frac{dT}{dt} = k(T - 75)$
 $T(t) = 340 \rightarrow 340 = 275e^{5k} + 75$
 $265 = 275e^{5k}$
 $\ln e^{5k} = \frac{265}{275}$
 $5k = \ln \frac{265}{275}$
 $k = -0.007$
 $T - 75 = \pm e^{kt}$
 $\rightarrow T(t) = (e^{kt} + 75)$
 $\rightarrow T(t) = 275e^{-0.007t} + 75 = 300$
 $275e^{-0.007t} = 225$
 $\ln e^{-0.007t} = \ln \frac{225}{275}$
 $-0.007t = \ln \frac{225}{275}$
 $t = 27$
 27-5 = 22 more min